

DBMS Lab 0

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Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

Q 1 What is SQL and its Use?

SQL is a standard language for accessing and manipulating databases.

SQL stands for Structured Query Language, which lets you access and manipulate data. It can perform CRUD operations i.e. Create, Read, Update and Delete the Data. One can create multiple tables with different data types in separate rows and columns. It can create various views in the database and can set permissions on the tables, procedures and views.

RDBMS (Relational Data Base Management System) is the basis of SQL and for all other modern database systems such as MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access.

Q 2 Describe all the different commands of SQL, mention use of it and syntax.

SQL allows various actions which are acted upon via the use of SQL statements. SQL statements contain basic keywords and end with a semi colon ";".

Here are some examples of the keywords used in SQL :-

- SELECT - extracts data from a database

SELECT CustomerName, City FROM Customers

- UPDATE - updates data in a database

UPDATE table name SET column1 = value1, column2 = value2 WHERE condition;

- DELETE - deletes data from a database

DELETE FROM table name WHERE condition;

. INSERT INTO - inserts new data into a database

INSERT INTO table name VALUES (value1, value2, value3,
...);

. CREATE DATABASE - creates a new database

CREATE DATABASE database_name;

. ALTER DATABASE - modifies a database

ALTER DATABASE OldDatabaseName MODIFY NAME
NewDatabaseName;

. CREATE TABLE - creates a new table

CREATE TABLE table_name (column1 datatype constraint,
column2 datatype constraint, column3 datatype constraint);

. ALTER TABLE - modifies a table

ALTER TABLE table name ADD column_name datatype;

. DROP TABLE - deletes a table

DROP TABLE table_name;

. CREATE INDEX - creates an index (search key)

CREATE INDEX index_name ON table_name (column1, column2, ...);

. DROP INDEX - deletes an index

DROP INDEX table_name.index_name;

Q 3 Write a step by step procedure to install MYSQL.

1. Download the MySQL Installer from the web ..

General Availability (GA) ReleasesArchives

MySQL Installer 8.0.46

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Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:

8.0.46

Select Operating System:

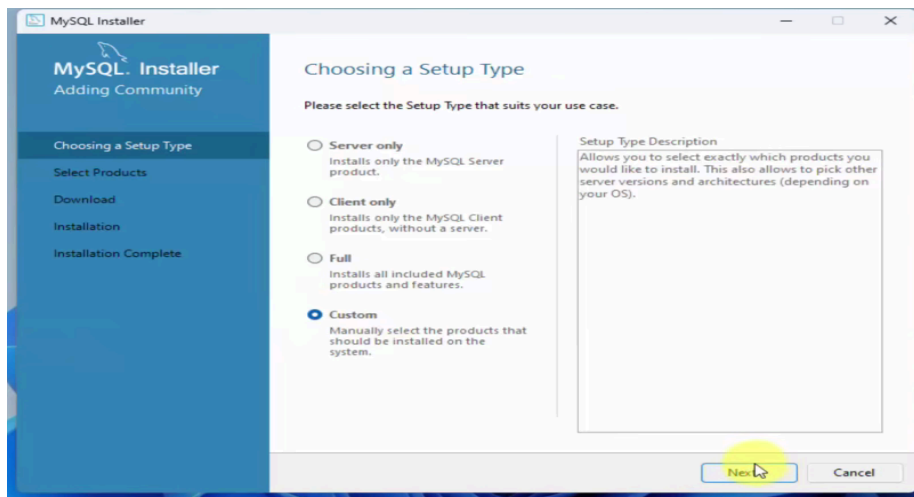
Microsoft Windows

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.46.0.msi)	8.0.46	2.1M	Download
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.46.0.msi)	8.0.46	565.9M	Download

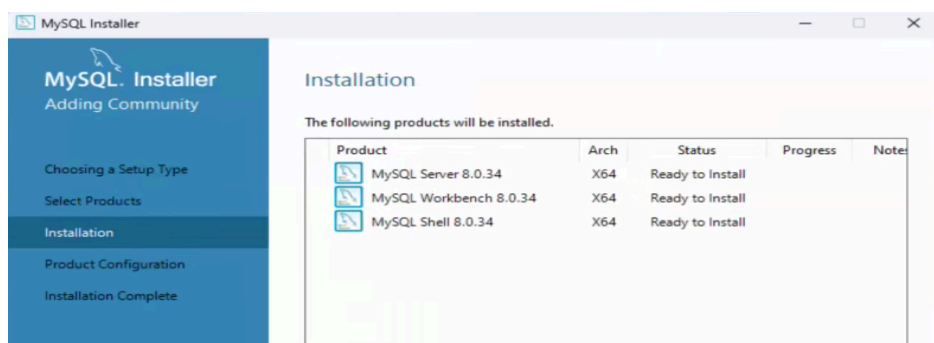
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We suggest that you use the [MD5 checksums](#) and [GnuPG signatures](#) to verify the integrity of the packages you download.

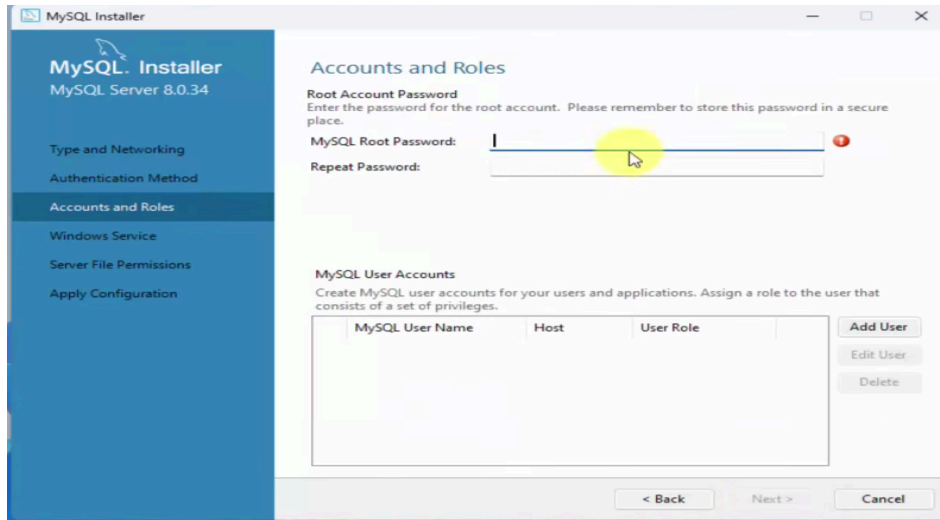
2. Run the installer and select the required setup type from the download window.



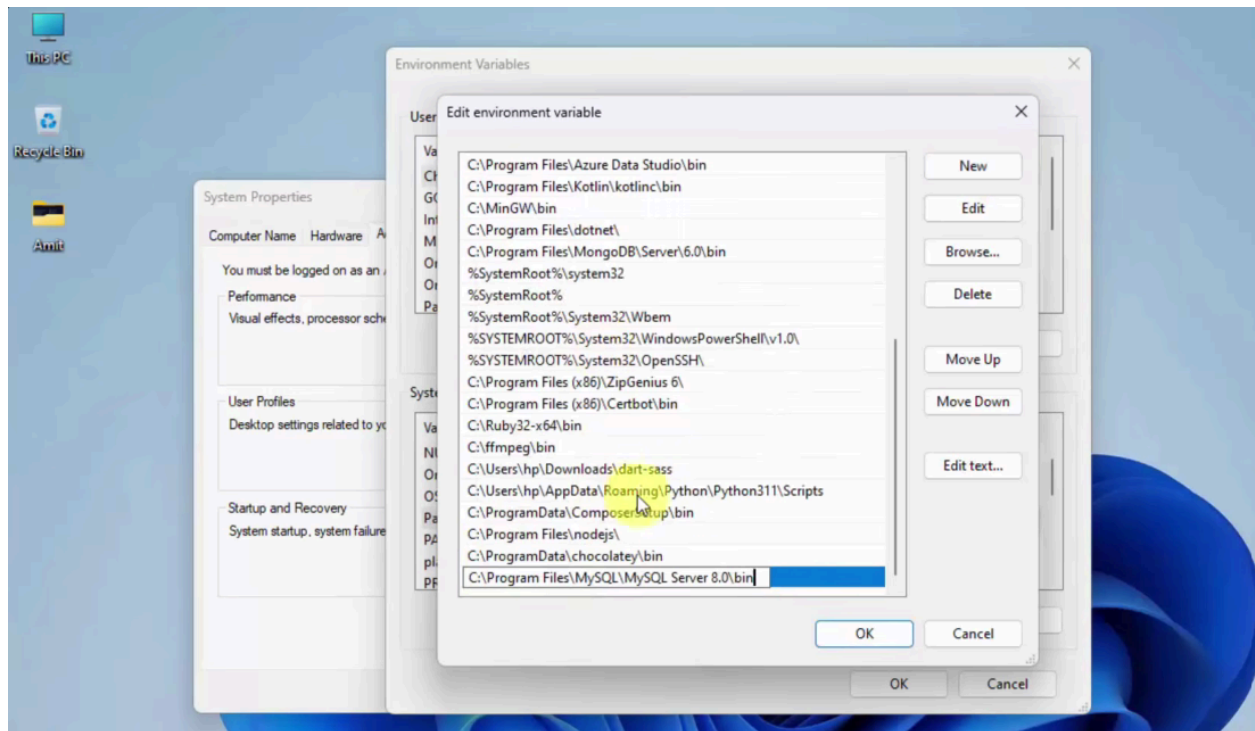
3. Install the required MySQL components.



4. Configure MySQL Server.

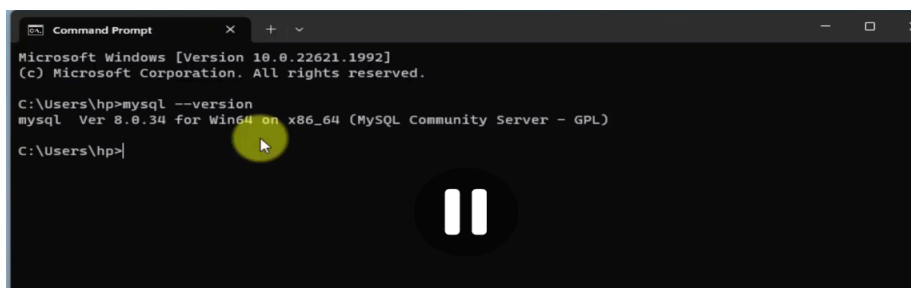


5. Set a password for the root user.



6. Complete the configuration process.

7. Open MySQL Workbench.



8. Connect using the MySQL credentials.

9. Verify installation using: `SHOW DATABASES;`

If databases are displayed, MySQL installation is successful.